

Math 142, F19
Quiz 5

Name: Answer

Instructions: You must show all of your work to receive credits for correct answers.

Problem 1. (6 points) Determine whether the following improper integrals are convergent or divergent and evaluate those that are convergent.

$$\begin{aligned} (a) \int_1^{+\infty} e^{-x/3} dx &= \lim_{b \rightarrow +\infty} \int_1^b e^{-x/3} dx \\ &= \lim_{b \rightarrow +\infty} \left[-3e^{-x/3} \right]_1^b \\ &= \lim_{b \rightarrow +\infty} -3(e^{-b/3} - e^{-1/3}) \\ &= -3(0 - e^{-1/3}) \\ &= \frac{3}{\sqrt[3]{e}} \end{aligned}$$

$$\begin{aligned} (b) \int_{-\infty}^{\pi} \cos(2\theta) d\theta &= \lim_{a \rightarrow -\infty} \int_a^{\pi} \cos(2\theta) d\theta \\ &= \lim_{a \rightarrow -\infty} \frac{1}{2} [\sin 2\theta]_a^{\pi} \\ &= \lim_{a \rightarrow -\infty} \frac{1}{2} [\sin 2\pi - \sin 2a] \\ &= \lim_{a \rightarrow -\infty} \left(-\frac{1}{2} \sin 2a \right) \rightarrow \text{limit doesn't exist!} \\ &\Rightarrow \text{divergent} \end{aligned}$$

Problem 2. (4 points) Evaluate the improper integral $\int_1^3 \frac{1}{\sqrt{x-1}} dx$ if possible.

$$\begin{aligned} \frac{1}{\sqrt{x-1}} &\text{ is discontinuous at } x=1 \\ \text{Thus} \quad \int_1^3 \frac{1}{\sqrt{x-1}} dx &= \lim_{a \rightarrow 1^+} \int_a^3 \frac{1}{\sqrt{x-1}} dx = \lim_{a \rightarrow 1^+} (2\sqrt{x-1}) \Big|_a^3 \\ &= \lim_{a \rightarrow 1^+} 2(\sqrt{2} - \sqrt{a-1}) \\ &= 2(\sqrt{2} - 0) = 2\sqrt{2} \end{aligned}$$